

EVOLUTION AND DIVERSITY

Until the 19th century, when a number of theories were proposed, it was a matter of speculation as to how such remarkably diverse life forms had developed on Earth. Today, the theory of evolution and diversification, alongside geological evidence for changes in the distribution of continents, give a fascinating insight into the ever-changing life on our planet.

CHANGE OVER TIME

All living things can change and adapt to their surroundings. Changes that are passed from generation to generation are hard to see, but over time they can alter the way a species looks or behaves. This process, known as evolution, is not gradual but has seen significant extinctions and bursts of growth over time.

The study of fossils to unravel the history of life was in its early stages in Charles Darwin's day, and since then a vast amount of information supporting the theory of evolution has emerged. We now know that life evolved in the oceans at least 3.7 billion years ago, and that it was from these early simple life forms that all current life on Earth evolved – including plants, fungi, and animals.

As life forms became more complex and moved from sea to land, the first forests and land-living animals evolved. The Mesozoic Era, which began 252 million years ago, with its successions of evolving plants and animals, produced the dominant dinosaur reptiles and their bird descendants. A mass extinction event 66 million years ago ushered in the Cenozoic Era. The reptiles were largely replaced by mammals both in the seas and on land, where flowering plants and their pollinating insects also became abundant and diverse.



< GIANT SALAMANDER

This extremely rare fossil skeleton of an *Andrias* (giant salamander) was mistaken for a human victim of the biblical Flood until French anatomist Georges Cuvier identified it as an amphibian in 1812.

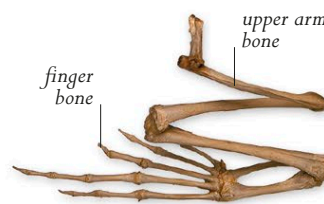
EVIDENCE OF EVOLUTION

Comparison of the anatomy of vertebrate limb bones from different species show that, despite different appearances and functions, they derive from the same basic developmental plan and the same genes.



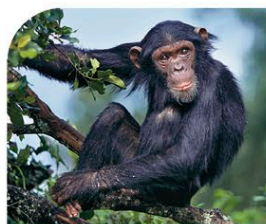
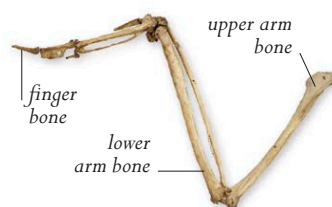
FROG

The frog's leg, arm, and finger bones are modified for swimming. Large muscles enable it to jump powerfully – essential for catching prey and escaping from predators.



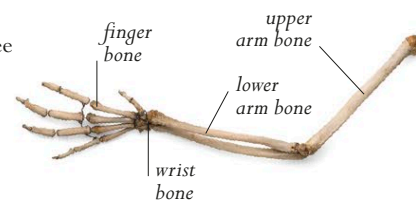
OWL

The wing of a bird is powered by flight muscles attached to the upper arm and bones of the wrist, with greatly modified and extended fingers.



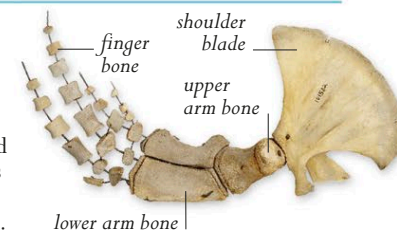
CHIMPANZEE

The arm of a chimpanzee is anatomically very similar to our own, but has slightly different proportions, with elongated fingers and a short thumb.



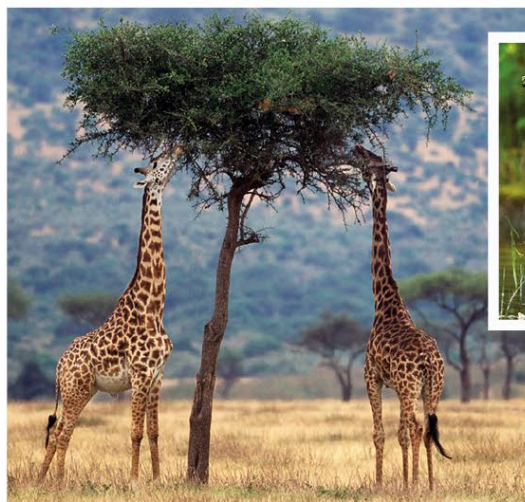
DOLPHIN

The arm bones of whales and dolphins form a flipper – with shortened, flattened, and strengthened arm bones and greatly lengthened second and third fingers.



LAMARCK LEADS THE WAY

The 18th-century French biologist Jean-Baptiste Lamarck developed the first comprehensive evolutionary theory that higher forms of life had “evolved” from simpler organisms. Based on his extensive research of invertebrates in particular, Lamarck argued that necessary characteristics could be acquired during an organism's lifetime – through the desire to gain food, shelter, and mates – and those not needed could be lost, with the resulting transformation passed on to its offspring. Although modern genetics disproves this theory of “soft inheritance”, Lamarck's concepts were an important starting point, and were developed further by the Scottish anatomist Robert Grant, who tutored Charles Darwin in Edinburgh. Darwin himself did not entirely rule out a Lamarckian mechanism, which he thought might be supplementary to natural selection.



< ^ INNER STRIVING

Lamarck believed that evolution worked through a process of “inner striving”. Thus the giraffe developed a long neck to reach the leaves of trees, and the heron grew long legs in order to wade.